8 bit Subtraction

AIM:

To write an assembly level language program to implement 8 bit subtraction using 8085 processor.

ALGORITHM:

1) Start the program by loading the first data into the accumulator.

2) Move the data to a register.

3) Get the second data and load it into the accumulator.

4) Subtract the two register contents.

5) Check for borrow.

6) Store the difference and borrow in the memory location.

7) Halt.

PROGRAM:

MNEMONICS	EXPLANATION
LDA 8000	Load accumulator with the first number in the address.
MOV B, A	Move the data from accumulator to B register.
LDA 8001	Load accumulator with the second number in the address.

SUB B	subtract the data B register with accumulator.
STA 8002	Store the data (output) of the accumulator in the address.
RST 1	Halt.

INPUT:

ADDRESS	DATA
8000	4
8001	5

OUTPUT:

ADDRESS	DATA
8002	1

O/P

RESULT:

Thus the program was executed successfully using 8085 processor simulator.

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

A	01	
BC	04	00
DE	00	00
HL	00	00
PSW	00	00
PC	42	0C
SP	FF	FF
Int-Reg	00	

Flag

S	0
Z	0
AC	0
P	0
C	0

Decimal - Hex Convertion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 − + 00

Update Port Value

Memory

8001 − + 05

Update Memory

Load me at

```
LDA 0000
MOV B,A
LDA 8001
SUB B
STA 8002
RST 1
```

Data Stack KeyPad **Memory** I/O Ports

Start 8002 OK

Address (Hex)	Address	Data
1F42	8002	1
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0
1F4E	8014	0
1F4F	8015	0
1F50	8016	0
1F51	8017	0

Line No	Assembler Message
0	Program assembled successfully

Simulator: Idle